Isometric Standard Project Verification Report

For

Charm Industrial

Client	Isometric
Company / GHG Removal Supplier	Charm Industrial
Facility or Project Name	Charm Kansas Bio-Oil Geologic Storage
Facility Address	**Bio-oil Production:** **AE Cote-Nord Canada Bioenergy Inc (AECN)** Port Cartier, Quebec, Canada **Charm Miniforge** Fort Lupton, CO, USA **Bio-Oil Pre-Processing:** **Charm Industrial** El Dorado, KS, USA **Bio-oil Injection:** **Vaulted Deep** Hutchinson, KS, USA
Greenhouse Gas Statement ID	Isometric ggs_1JV6H8C4F1S0CY5F
Net Greenhouse Gas Removals	1002.95 tonnes
Removal Method	Bio-Oil Geological Storage
Removal Period	Feb 1, 2025 – Mar 31, 2025
Verification Opinion	Fully meets criteria
Verifier/Validation and Verification Body	Tim Hansen 350Solutions, Inc.

ISSUED: JUNE 5, 2025

CONTENTS

1. Key Project Data 3
1.1. Verification Summary 4
2. Verification Details 6
2.1. Verification Objectives 6
2.2. Level of Assurance & Materiality 6
2.3. Verification Team Details 6
2.4. Audit Process Followed 6
3. Technical Aspects 7
3.1. Project Details 7
3.2. Protocols and Modules Applied 8
3.3. Applicability of Applied Protocol and Modules 8
3.4. Project Boundary 8
3.5. Changes from Previous Verifications 9
3.6. Process & Data Irregularities During Reporting Period 9
4. Crediting Details 10
4.1. Project Reporting Period 10
4.2. Crediting Period 10
5. GHG Statement Verification 10
5.1. Accuracy of the Statement 10
5.2. Verification Opinion 11
5.3. Verification Results Summary 11
5.4. Verification Findings 14
6. Revision History 15
Appendix 1: Documents Submitted for Review 17
Appendix 2: Verification Audit Findings and Responses 31
Appendix 3: Verifier Qualifications 36

ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Kelly Inder-Nesbitt
Removal Method: Bio-Oil Geological Storage		
Report Date: June 5, 2025		
Document No: 350VR-Charm-ISO2504.03		
Rev: v1.2		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14 Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	February 1, 2025 – March 31, 2025
Greenhouse Gas Statement ID	Isometric ggs_1JV6H8C4F1S0CY5F (Statement ID) Isometric gss_1JV6H8TS01S00K14 (Version ID)
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	1002.95 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns
Verification Report Date	June 5, 2025

Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of carbon removal credits by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.7 • Bio-Oil Geological Storage Protocol v1.1 • Biomass Feedstock Accounting Module v1.2 • Biomass or Bio-oil Storage in Salt Caverns v1.1 • Embodied Emissions Accounting v 1.0 • Energy Use Accounting v 1.2 • Transportation Emissions Accounting v1.1

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period including February 1, 2025 – March 31, 2025. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Kansas Bio-oil Geologic Storage project for use of bio-oil produced by Charm was completed on October 18, 2024 for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING MARCH 31, 2025

Removal ID	Supplier reference ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1JV2X2DS51S0K3AK	782	2/3/2025	2/3/2025	22030
rmv_1JV5N6F0J1S0VZ2N	781	2/5/2025	2/5/2025	48220
rmv_1JTRDZTAA1S079AP	784	2/7/2025	2/7/2025	47910
rmv_1JTRE7K7Q1S0V5SD	786	2/10/2025	2/10/2025	47540
rmv_1JV2R5WYE1S0T83W	783	2/17/2025	2/17/2025	46580
rmv_1JTRF1BHP1S0M4DF	788	2/24/2025	2/24/2025	48520
rmv_1JV2QKMM41S09YNS	785	3/3/2025	3/3/2025	45460
rmv_1JTRFB2HV1S07D5Z	790	3/5/2025	3/5/2025	45400
rmv_1JV2ED0K81S07JNE	787	3/7/2025	3/7/2025	43380
rmv_1JTRG60TD1S0JAP7	792	3/10/2025	3/10/2025	46630

rmv_1JV2F6WKN1S0BT4K	789	3/11/2025	3/11/2025	48970
rmv_1JV3AP01P1S034QR	791	3/13/2025	3/13/2025	41730
rmv_1JTRGGJTD1S006HM	794	3/14/2025	3/14/2025	43710
rmv_1JTRGSDK71S0KFA4	796	3/17/2025	3/17/2025	46980
rmv_1JV2RG2AJ1S04436	793	3/18/2025	3/18/2025	57090
rmv_1JV2RR1B31S0S7B8	795	3/20/2025	3/20/2025	45010
rmv_1JTRH0XBT1S0SNJ0	798	3/21/2025	3/21/2025	48520
rmv_1JV2S8S941S0TZNC	797	3/24/2025	3/24/2025	47450
rmv_1JTRSV5621S0SDR0	800	3/25/2025	3/25/2025	47500
rmv_1JV2YJPQX1S0NKM3	799	3/27/2025	3/27/2025	44250
rmv_1JV2QSR181S0JX15	801	3/28/2025	3/28/2025	44540
rmv_1JV2RX3TP1S0532H	803	3/31/2025	3/31/2025	45530
TOTAL Verified Removals				**1002950 kg** **1002.95 tonnes**

The quantity of 1002.95 tonnes of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2.1*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VR-Charm-ISO2504.03

Timothy A. Hansen — Digitally signed by Timothy A. Hansen Date: 2025.06.05 22:24:18 -04'00'

Signed: Tim Hansen (Verifier)

Kelly Inder-Nesbitt — Digitally signed by Kelly Inder-Nesbitt Date: 2025.06.05 22:29:23 -04'00'

Kelly Inder-Nesbitt (QA)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the time-period February 1, 2025 – March 31, 2025. The verification was conducted following the specifications of *Isometric Standard V1.7, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*. The verification is based on the provision and fair presentation of a complete and accurate GHG Statement by Charm Industrial and Isometric. This is confirmed and verified by 350Solutions based on supporting evidence in accordance with the Isometric criteria (Table 1, Section 3.2).

2.2. LEVEL OF ASSURANCE & MATERIALITY

The verification was conducted with a reasonable level of assurance. Any discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.3. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020 for completion of ISO 14034 Technology Verifications. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was provided QA by Kelly Inder-Nesbitt, a Senior Carbon Removal Verification Engineer, with experience developing MRV requirements for and verifying carbon removal technologies. Complete verifier qualifications are attached in Appendix 3.

2.4. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil Geologic storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
May 14-June 4, 2025	Review of reporting period modifications and changes	- Review of GHG Statement and report and guidance from Charm	PDD (V 1.9) Feb_Mar Removals Isometric GHG Statement Report v1.0 Verifier Guidance_ Feb_Mar 2025 Isometric ggs_1JV6H8C4F1S0CY5F (Statement ID) Isometric gss_1JV6H8TS01S00K14 (Version ID)

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
June 1 – June 5, 2025	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations and changes in RP - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues and findings and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 ISO2504_Charm_RP7Dec24-31Jan25_VR_V1.2FINAL_16APR2025 PDD (V 1.9) Feb_Mar Removals Isometric GHG Statement Report v1.0 Verifier Guidance_ Feb_Mar 2025 Isometric ggs_1JV6H8C4F1S0CY5F (Statement ID) Isometric gss_1JV6H8TS01S00K14 (Version ID) Isometric Standard V1.4, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.2
June 1 – June 5, 2025	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information - Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtained bio-oil from two sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. In addition, Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Bio-oil from both facilities is shipped by tanker truck or rail car to a Charm facility in El Dorado, Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection

permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored on site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are detailed in their own PDD, which has been previously validated [1] and remains valid for the reporting period.

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.7 [2]
- Bio-Oil Geological Storage Protocol v1.1 [3]
- Biomass Feedstock Accounting Module v1.2 [4]
- Biomass or Bio-oil Storage in Salt Caverns v1.1 [5]
- Embodied Emissions Accounting v 1.0 [6]
- Energy Use Accounting v 1.2 [7]
- Transportation Emissions Accounting v 1.1 [8]

Transportation Emissions Accounting Module v1.0 was used in the initial validation and previous verifications of this project. The module was updated (V1.1) to allow for the use of Book and Claim Units (BCUs) to substitute for some, or all, or project fuel usage.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into salt caverns. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply of wood waste to the biomass processing sites (AECN and Charm Fort Lupton), the conversion of the biomass to bio-oil via AECN's pyrolysis process and Charm's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made modifications to the process, data collection, or calculation procedures which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period:

- New equipment was added at Charm's facilities:
    - A full list of additional equipment purchased and installed, as well as components of a stormwater pond were provided
    - For those equipment items for which carbon intensity (embodied emissions) had not previously been established, cradle to grave emission factors were established and details and justification provided for sourcing from quality and conservative sources (i.e. Ecoinvent)
    - Embodied emissions were amortized and allocated using compliant procedures beginning the first day of the reporting period (Feb 1).
- Charm completed an annual emission factor review and update, which resulted in some emission factor changes (i.e. GREET 2024, GLEC v3.1).
- An emissions test on the emissions control device for the Charm Fort Lupton pyrolysis operations was completed in late October 2024. This reporting period is the first where this new emissions data was utilized to calculate emissions of CO_2, CH_4, and N_2O from the pyrolysis process tail gas.
- Charm modified the methods for computing the amount of salt required and the mass of oil injected to allocate upstream production and transport emissions to removals. The procedure was reviewed by Isometric and 350Solutions and found to be acceptable and compliant with protocols.
- Charm was modifying its approach for reporting Charm bio-oil water content throughout the period. Initially, Charm was estimating water content at 30%, conservatively, based on historical averages of 25%. Going forward, Charm is using the average of water content analytical results for each batch. During this transition, some procedures were not fully followed that resulted in modified calculations. However, these were reviewed and determined to overestimate water content, resulting in higher salt additions than required and no compliance issues with injectate salt requirements.
- Charm has discontinued the use of containers on flatbed trucks for transporting bio-oil-filled totes. As a result, the truck mass calculation is now simplified, requiring only the gross weight of the full totes for transportation emissions accounting

3.6. PROCESS & DATA IRREGULARITIES DURING REPORTING PERIOD

Several anomalies and irregularities occurred during the reporting period related to data collection and operations. All have been reviewed and the approach Charm has taken for each accepted and determined to be compliant with protocols. The following issues were identified by Charm and reviewed by 350Solutions:

- Charm performed tank cleaning during the period that resulted in additional use of water as well as requiring disposal of the salt contaminated water (disposed on site using a permit disposal well). Charm documented all water purchases, shipments, and disposal and properly accounted for all activities.

- Four batches of bio-oil blends for injection were sampled in a different manner than planned in the PDD. For these batches, carbon content was determined using weighted averages of production batch carbon contents to determine the blended batch C content. This is acceptable in accordance with the protocol and was previously approved by Isometric.
- A water flow meter for recording water consumption became inoperable during a brief period during production. For those periods, water consumption is reported as 4 gallons per hour during production 'hot flow' periods. Cross checks vs. historical meter data are made to ensure this assumption is conservative and metering data resumed as the primary measure beginning Feb 21.
- For a salt delivery on 3/28, the empty truck weight was not recorded. Charm utilized the lowest previously recorded empty truck weight from prior deliveries as a substitute data point. This is considered conservative and appropriate by 350.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil by Charm during the period February 1, 2025 to March 31, 2025. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period, but properly accounted for and allocated to the bio-oil injected within the reporting period.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024 and continues for five years, ending February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Review and confirmation of calculations of total bio-oil and CO_2 equivalent injected as well as all supporting evidence for the following removals

rmv_1JV2X2DS51S0K3AK	**782**	**2/3/2025**
rmv_1JV2RG2AJ1S04436	**793**	**3/18/2025**
rmv_1JV2RX3TP1S0532H	**803**	**3/31/2025**

- This includes:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS))
    - Review and calculation of average carbon content from raw data (lab reports)
    - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS

  - Confirmation of weigh scale raw data for deliveries
  - Re-calculation of transportation associated emissions for such transport.
  - Confirmation of raw data for fuel usage at Charm pre-treatment facility
  - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
  - Confirmation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage, respectively
  - Confirmation of updated emission factors for transportation, embodied emissions, and tailgas emissions
  - Review of Charm pyrolysis process emissions, supporting evidence, and calculations
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit.

As with the previous verification, some minor discrepancies were noted due to rounding errors both internally at Charm and in communication between Charm's and Isometric's data platforms. The estimated impact of the total discrepancies is not a material issue.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Meets the requirements with <u>minor revisions</u> of Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.7, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation
2.5.5	*Default Emission Factors, Proxies, and Models*	Y	Acceptable – Charm completed an annual update of emission factors and utilized appropriate updates and quality factors. With the addition of some new equipment, additional emission factors and calculations were used for new equipment embodied emissions. All calculations and factors comply with protocol requirements and are from high quality sources, with justification of selection provided. No additional significant changes were made since prior validations.
2.5.11	*Life Cycle Assessment (LCA) Policies*	Y	
2.5.6	*Common Calculation Factors*	N	
2.5.7	*Uncertainty*	N	
2.5.8	*Durability and Monitoring*	N	Acceptable – No change since validation
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	Y	Acceptable – Charm provided documentation of modifications to processes and calculation procedures, updated documentation on emission testing, and all other required documentation. No other changes since validation
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Acceptable – Maintenance of well operation permits meets requirements and ongoing permit reporting requirements completed.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since validation
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Integrity*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Acceptable - All required parameters measured and are within the required limits.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable – Brine TOC measured prior to emplacement, as required by Class V permit. Confirmed via updated TOC data.
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	NA	Acceptable - No change since validation
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	NA	Acceptable - No change since validation
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit (February and March monthly reports and Quarterly groundwater monitoring report to KDHE provided)
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
2.1	*Permit for Operation*	N	Acceptable – No change since validation
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.1	*Sampling Frequency*	N	Acceptable – No change since validation. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly. Updated monitoring data provided as required.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed.

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open Forward Action Requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

Type	Finding / Issue	Source	Protocol Reference	Action
Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing.	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	7.4.3.5	Please provide AECN testing report. Note, not applicable for this reporting period, but should be updated when possible
Forward Action Request	Please provide a copy of the next quarterly Report to KDHE submitted by Vaulted for Q12025 (emplaced material	NA	3.1.3	Provide copy of KDHE quarterly report on emplacement level

	level in cavern). This is not required for the current reporting period, but will be for the next.			
Forward Action Request	There have been some instances of higher C content than typically previously seen (>50%) in some batches now. This may be a sign of process improvements and achieving higher C content. However, please review batch carbon content history to ensure these are not outliers that could be the result of sample collection, mixing, or other systemic issues and that they are representative. If such samples are outliers, Charm could employ the outlier requirements in Section 8.2.1.1.1 of Protocol to address this as a conservative approach	Charm Blend Batch C Content.xls	8.2.1.1.1	Review historical C content trends for Charm bio-oil samples to determine mean and standard deviation for potential future evaluation of outliers

6. REVISION HISTORY

Original date of issue: June 5th, 2025

Version	Date Issued	Noted Changes
V1.0 (Draft – Internal)	June 4, 2025	N/A Initial Version
V1.1 (Draft – External)	June 5, 2025	Minor edits from QA
V1.2 (FINAL)	June 5, 2025	No changes

References

[1] "Vaulted Deep Great Plains Facility Organic Waste Sequestration: Isometric PDD." Jan. 23, 2025. Accessed: Feb. 19, 2025. [Online]. Available: https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H43SYY631S0Z8YR/prj_1HHYZFVGW1S044ZY/pjd_1JJSMS9FS1S0MF1M/pdd.pdf

[2] Isometric, "Isometric Standard v1.2.0." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[3] Isometric, "Bio-oil Geological Storage V1.1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[4] Isometric, "Biomass Feedstock Accounting V1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[5] Isometric, "Biomass or Bio-oil Storage in Salt Caverns V1.0.4." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[6] Isometric, "Embodied Emissions Accounting v1.0.3." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[7] Isometric, "Energy Use Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[8] Isometric, "Transportation Emissions Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

File Name	File Type
V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx	PDD
Feb_Mar Removals Isometric GHG Statement Report v1.0.docx	Additional Guidance and Documentation\
Verifier Guidance_ Feb_Mar 2025.docx	Additional Guidance and Documentation\
[External] Upstream Emissions Allocation Brief.docx	Additional Guidance and Documentation\
Charm Blend Batch C Content.xlsx	Charm BO MRV\Charm Blend Batch C Content\
WODO Tote samples for PP2.pdf	Charm BO MRV\Charm Blend Batch C Content\
Charm Deliveries Detail.xlsx	Charm BO MRV\Charm BO Deliveries\
1000_ Spool.docx	Charm BO MRV\Charm Production MRV Docs\New Capital + Equipment LCA Docs\
2,000 Gallon Ag Spray Tank.docx	Charm BO MRV\Charm Production MRV Docs\New Capital + Equipment LCA Docs\
250 Gallon Diesel Fuel Tank.docx	Charm BO MRV\Charm Production MRV Docs\New Capital + Equipment LCA Docs\
40_x8_ Conex Shipping Container.docx	Charm BO MRV\Charm Production MRV Docs\New Capital + Equipment LCA Docs\
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Emissions Calculations - Methane and N2O.xlsx	Charm BO MRV\Charm Production MRV Docs\October Tailgas Testing\
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PP4 Waste Disposal Emissions Calc Sheet.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 4\Waste Disposal\
PP4 Water Usage Log.xlsx	Charm BO MRV\Charm Production MRV Docs\Production Period 4\Water\
AECN Upstream Allocations.xlsx	Lot Data\
185-9896.pdf	Lot Data\2-75\
185-9898.pdf	Lot Data\2-75\
2-75 Ops Notes.pdf	Lot Data\2-75\
BATCHES 200074, 200075, 3-06 MRV (CHN + H2O) (4).pdf	Lot Data\2-75\
BATCHES 200074, 200075, 3-06 MRV (TAN + Density) (1).pdf	Lot Data\2-75\
185-9961.pdf	Lot Data\2-76\
185-9964.pdf	Lot Data\2-76\
2-76 MRV Ops Notes.pdf	Lot Data\2-76\
BATCH 200076 and 3-07 #1 MRV (TAN + Density).pdf	Lot Data\2-76\
BATCH 200076 and 3-07 #1 MRV(CHN + H2O).pdf	Lot Data\2-76\
185-9985.pdf	Lot Data\2-77\
185-9986.pdf	Lot Data\2-77\
2-77 MRV Ops Notes.pdf	Lot Data\2-77\
BATCH 200077 #1 MRV (TAN + Density).pdf	Lot Data\2-77\
BATCH 200077 #1 MRV(CHN + H2O).pdf	Lot Data\2-77\
185-10037.pdf	Lot Data\2-78\
185-10038.pdf	Lot Data\2-78\
2-78 MRV Ops Notes.pdf	Lot Data\2-78\
BATCH 3-08 3-09 2-78 (CHN & H2O).pdf	Lot Data\2-78\
BATCH 3-08 3-09 2-78 (TAN & Density).pdf	Lot Data\2-78\
185-10141.pdf	Lot Data\2-79\
185-10142.pdf	Lot Data\2-79\
2-79 MRV Ops Notes.pdf	Lot Data\2-79\
BATCH 2-79 #1 3-10 #1 (CHN & H2O).pdf	Lot Data\2-79\
BATCH 2-79 #1 3-10 #1 (TAN & Density).pdf	Lot Data\2-79\
185-10199.pdf	Lot Data\2-80\
185-10202.pdf	Lot Data\2-80\
2-80 MRV Ops Notes.pdf	Lot Data\2-80\
FINAL MRV 2-80 3-11 (TAN & Density).pdf	Lot Data\2-80\
FINAL SPARGE 2-80 & Charm AECN blend 3-11 (CHN & H2O).pdf	Lot Data\2-80\
185-10257.pdf	Lot Data\2-81\
185-10259.pdf	Lot Data\2-81\
2-81 MRV Ops Notes.pdf	Lot Data\2-81\
Batch 2-81 & 3-12 (TAN & Density).PDF	Lot Data\2-81\

Oil BATCH 2-81 & 3-12 (CHN & H2O).pdf	Lot Data\2-81\
185-10267.pdf	Lot Data\2-82\
185-10268.pdf	Lot Data\2-82\
2-82 MRV Ops Notes.pdf	Lot Data\2-82\
200082 & 300013 (CHN & H2O).pdf	Lot Data\2-82\
200082 & 300013 (TAN & Density).PDF	Lot Data\2-82\
185-10336.pdf	Lot Data\2-83\
185-10338.pdf	Lot Data\2-83\
2-83 MRV Ops Notes.pdf	Lot Data\2-83\
200083 & 300014 SR25-02097 BATCH FINAL MRV (TAN & Density).pdf	Lot Data\2-83\
200083 FINAL SPARGE & 300014 Charm AECN blend (CHN & H2O).pdf	Lot Data\2-83\
185-10372.pdf	Lot Data\2-84\
185-10373.pdf	Lot Data\2-84\
2-84 MRV Ops Notes.pdf	Lot Data\2-84\
200084 & 3000015 (CHN & H2O).PDF	Lot Data\2-84\
200084 & 300015 (TAN & Density).PDF	Lot Data\2-84\
.DS_Store	Lot Data\2-85\
185-10486.pdf	Lot Data\2-85\
185-10487.pdf	Lot Data\2-85\
2-85 MRV Ops Notes.pdf	Lot Data\2-85\
Batch 3-18 #1, Batch 2-85 #1 (CHN & H2O) (1).pdf	Lot Data\2-85\
Batch 3-18 #1, Batch 2-85 #1 (TAN & Density) (1).pdf	Lot Data\2-85\
185-9930.pdf	Lot Data\3-07\
185-9931.pdf	Lot Data\3-07\
3-07 Charm Contents.xlsx	Lot Data\3-07\
3-07 Ops Notes.pdf	Lot Data\3-07\
BATCH 200076 and 3-07 #1 MRV (TAN + Density).pdf	Lot Data\3-07\
BATCH 200076 and 3-07 #1 MRV(CHN + H2O).pdf	Lot Data\3-07\
185-10024.pdf	Lot Data\3-08\
185-10026.pdf	Lot Data\3-08\
3-08 Charm Contents.xlsx	Lot Data\3-08\
3-08 MRV Ops Notes.pdf	Lot Data\3-08\
BATCH 3-08 3-09 2-78 (CHN & H2O).pdf	Lot Data\3-08\
BATCH 3-08 3-09 2-78 (TAN & Density).pdf	Lot Data\3-08\
185-10117.pdf	Lot Data\3-09\
185-10118.pdf	Lot Data\3-09\
3-09 MRV Ops Notes.pdf	Lot Data\3-09\
BATCH 3-08 3-09 2-78 (CHN & H2O).pdf	Lot Data\3-09\
BATCH 3-08 3-09 2-78 (TAN & Density).pdf	Lot Data\3-09\
Charm Contents 3-09.xlsx	Lot Data\3-09\

185-10185.pdf	Lot Data\3-10\
185-10186.pdf	Lot Data\3-10\
3-10 Charm Contents.xlsx	Lot Data\3-10\
3-10 MRV Ops Notes.pdf	Lot Data\3-10\
BATCH 2-79 #1 3-10 #1 (CHN & H2O).pdf	Lot Data\3-10\
BATCH 2-79 #1 3-10 #1 (TAN & Density).pdf	Lot Data\3-10\
185-10217.pdf	Lot Data\3-11\
185-10218.pdf	Lot Data\3-11\
3-11 Charm Contents.xlsx	Lot Data\3-11\
3-11 MRV Ops Notes.pdf	Lot Data\3-11\
FINAL MRV 2-80 3-11 (TAN & Density).pdf	Lot Data\3-11\
FINAL SPARGE 2-80 & Charm AECN blend 3-11 (CHN & H2O).pdf	Lot Data\3-11\
185-10246.pdf	Lot Data\3-12\
185-10247.pdf	Lot Data\3-12\
3-12 Charm Contents.xlsx	Lot Data\3-12\
3-12 MRV Ops Notes.pdf	Lot Data\3-12\
Batch 2-81 & 3-12 (TAN & Density).PDF	Lot Data\3-12\
Oil BATCH 2-81 & 3-12 (CHN & H2O).pdf	Lot Data\3-12\
185-10284.pdf	Lot Data\3-13\
185-10285.pdf	Lot Data\3-13\
200082 & 300013 (CHN & H2O).pdf	Lot Data\3-13\
200082 & 300013 (TAN & Density).PDF	Lot Data\3-13\
3-13 Charm Contents.xlsx	Lot Data\3-13\
3-13 MRV Ops Notes.pdf	Lot Data\3-13\
185-10314.pdf	Lot Data\3-14\
185-10315.pdf	Lot Data\3-14\
200083 & 300014 SR25-02097 BATCH FINAL MRV (TAN & Density).pdf	Lot Data\3-14\
200083 FINAL SPARGE & 300014 Charm AECN blend (CHN & H2O).pdf	Lot Data\3-14\
3-14 Charm Contents.xlsx	Lot Data\3-14\
3-14 MRV Ops Notes.pdf	Lot Data\3-14\
185-10353.pdf	Lot Data\3-15\
185-10354.pdf	Lot Data\3-15\
200084 & 3000015 (CHN & H2O).PDF	Lot Data\3-15\
200084 & 300015 (TAN & Density).PDF	Lot Data\3-15\
3-15 Charm Contents.xlsx	Lot Data\3-15\
3-15 MRV Ops Notes.pdf	Lot Data\3-15\
185-10411.pdf	Lot Data\3-16\
185-10412.pdf	Lot Data\3-16\
3-16 Charm Contents.xlsx	Lot Data\3-16\
3-16 MRV Ops Notes.pdf	Lot Data\3-16\

CHARM 3-17 and 3-16 (CHN & H2O).PDF	Lot Data\3-16\
CHARM 3-17 and 3-16 (TAN & Density).pdf	Lot Data\3-16\
185-10435.pdf	Lot Data\3-17\
185-10436.pdf	Lot Data\3-17\
3-17 Charm Contents.xlsx	Lot Data\3-17\
3-17 MRV Ops Notes.pdf	Lot Data\3-17\
CHARM 3-17 and 3-16 (CHN & H2O).PDF	Lot Data\3-17\
CHARM 3-17 and 3-16 (TAN & Density).pdf	Lot Data\3-17\
185-10459.pdf	Lot Data\3-18\
185-10460.pdf	Lot Data\3-18\
3-18 Charm Contents.xlsx	Lot Data\3-18\
3-18 MRV Ops Notes.pdf	Lot Data\3-18\
Batch 3-18 #1, Batch 2-85 #1 (CHN & H2O) (1).pdf	Lot Data\3-18\
Batch 3-18 #1, Batch 2-85 #1 (TAN & Density) (1).pdf	Lot Data\3-18\
.DS_Store	Lot Data\AECN BOLs\
Biling document 1949.pdf	Lot Data\AECN BOLs\
Biling documents 1937.pdf	Lot Data\AECN BOLs\
Billing Camion 2074-T.pdf	Lot Data\AECN BOLs\
Billing Doc 2005-T Corr.pdf	Lot Data\AECN BOLs\
Billing Docs 1934 - CORRECTIONS.pdf	Lot Data\AECN BOLs\
Billing Docs 1972-T.pdf	Lot Data\AECN BOLs\
Billing docs 2027-T Corrections.pdf	Lot Data\AECN BOLs\
Billing Docs 2036-T (corrections).pdf	Lot Data\AECN BOLs\
Billing Docs 2082-T - Corr Empty wt, and total loaded.pdf	Lot Data\AECN BOLs\
Billing document 2068-T.pdf	Lot Data\AECN BOLs\
Billing document 2071-T.pdf	Lot Data\AECN BOLs\
Billing document 2072-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1917.pdf	Lot Data\AECN BOLs\
Billing Documents 1928.pdf	Lot Data\AECN BOLs\
Billing Documents 1929.pdf	Lot Data\AECN BOLs\
Billing Documents 1930.pdf	Lot Data\AECN BOLs\
Billing Documents 1954-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1956-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1963-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1964-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1967-T.pdf	Lot Data\AECN BOLs\
Billing Documents 1997-T.pdf	Lot Data\AECN BOLs\
Billing documents 2008.pdf	Lot Data\AECN BOLs\
Billing Documents 2010-T.pdf	Lot Data\AECN BOLs\
Billing Documents 2022-T.pdf	Lot Data\AECN BOLs\
Billing documents 2028-T.pdf	Lot Data\AECN BOLs\

Billing Documents 2029.pdf	Lot Data\AECN BOLs\
Billing Documents 2038-T.pdf	Lot Data\AECN BOLs\
Billing documents 2049-T.pdf	Lot Data\AECN BOLs\
Billing documents 2058-T - Corrections.pdf	Lot Data\AECN BOLs\
Billing Documents 2066-T - Corrections.pdf	Lot Data\AECN BOLs\
Billing Documents 2079-T.pdf	Lot Data\AECN BOLs\
Billing Documents 2083-T.pdf	Lot Data\AECN BOLs\
Billing documents 2085-T.pdf	Lot Data\AECN BOLs\
Docs Truck 2056-T - Corrections.pdf	Lot Data\AECN BOLs\
documents charm 1974-T.pdf	Lot Data\AECN BOLs\
Envoi des documents Camion 2015-T.pdf	Lot Data\AECN BOLs\
Envoi des documents Camion 1948.pdf	Lot Data\AECN BOLs\
Envoi des documents Camion 2019-T.pdf	Lot Data\AECN BOLs\
Envoi des documents Camion 2063-T.pdf	Lot Data\AECN BOLs\
RS1-250116-1 Final CHARM .pdf	Lot Data\AECN BOLs\COAs\
RS1-250127-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250205-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250211-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250225-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250303-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250307-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250310-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250314-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250318-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS1-250323-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
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RS2-250202-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS3-250111-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS3-250123-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS3-250208-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
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RS3-250223-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
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RS3-250305-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS3-250316-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
RS3-250320-1 Final CHARM.pdf	Lot Data\AECN BOLs\COAs\
Billing Documents 1902.pdf	Lot Data\AECN BOLs\Partials\
Docs Charm 1910 CORRECTIONS.pdf	Lot Data\AECN BOLs\Partials\
documents charm 1915.pdf	Lot Data\AECN BOLs\Partials\
cHARM.pdf	Site Emissions\Diesel\
Charm_cWUKsvt.pdf	Site Emissions\Diesel\

CHARM_FUpPYEu.pdf	Site Emissions\Diesel\
CHARM_rQrDknc.pdf	Site Emissions\Diesel\
Feb_March Diesel Consumption Calcs.xlsx	Site Emissions\Diesel\
252799599470.pdf	Site Emissions\Electricity\Minifinery\
bill_252641473420 (2).pdf	Site Emissions\Electricity\Minifinery\
Minifinery Feb_March Consolidated Energy Consumption.xlsx	Site Emissions\Electricity\Minifinery\
Vaulted Consolidated Electricity Consumption.xlsx	Site Emissions\Electricity\Vaulted\
185-10024 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10026 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10037 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10038 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10117 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10118 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
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185-10142 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10185 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10186 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10199 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10202 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10217 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10218 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10246 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10247 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10257 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10259 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
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185-10284 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10285 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10314 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10315 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10336 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10338 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10353 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10354 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10372 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10373 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10411 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10412 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10435 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10436 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\

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185-10486.pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-10487.pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-9896.pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-9898 (2).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-9930 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
185-9931 (1).pdf	Site Emissions\Electricity\Vaulted\Vaulted IE Docs\
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0052860440.PDF	Site Emissions\LCS\
0052860441.PDF	Site Emissions\LCS\
0052885912.PDF	Site Emissions\LCS\
0052898707.PDF	Site Emissions\LCS\
Feb_March LCS Consumption Sheet.xlsx	Site Emissions\LCS\
Copy of 0.jpg	Site Emissions\Salt Deliveries\
Copy of 2_6 Salt Delivery.pdf	Site Emissions\Salt Deliveries\
Copy of 2_7 Salt Delivery.pdf	Site Emissions\Salt Deliveries\
Copy of 4575_001.pdf	Site Emissions\Salt Deliveries\
Feb_Mar Salt Deliveries Calc Sheet.xlsx	Site Emissions\Salt Deliveries\
.DS_Store	Site Emissions\Waste Disposal\
Minifinery to Water Station.png	Site Emissions\Waste Disposal\Tank Cleanouts\
Tank Cleanouts Calc Sheet Feb_Mar.xlsx	Site Emissions\Waste Disposal\Tank Cleanouts\
Transwood to Minifinery.png	Site Emissions\Waste Disposal\Tank Cleanouts\
Invoice37747.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
Invoice37764.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
Invoice37877.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
invoice37933.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
invoice37959 (1).pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
Invoice47393.pdf	Site Emissions\Waste Disposal\Tank Cleanouts\Cleanout Invoices\Feb-March\
Feb TOC.pdf	Vaulted MRV\Brine TOC Reports\
Mar TOC.pdf	Vaulted MRV\Brine TOC Reports\
2025 1st Quarter Monitoring Well Report.pdf	Vaulted MRV\Vaulted Reports\
AR-09 Feb 2025 Report.pdf	Vaulted MRV\Vaulted Reports\
AR-09 KOLAR March 2025.pdf	Vaulted MRV\Vaulted Reports\

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Audit Evidence	Protocol Reference	Required Action	Supplier Response	Conclusion / Resolution
1	Clarification	Certificate of Analysis: SR25-02319.001 indicates "# - Result is outside of test method limits and/or analytical range used in method precision study" for the carbon content test. Although this is not likely an issue, can Charm request clarification from SGS regarding what this means and what the range for the method precision study was?	WODO Tote samples for PP2 and other COAs	8.2.1.1.1	Provide clarification and supporting evidence regarding instrument calibration for C content analyses by SGS and that analytical results are within calibration ranges	From SGS: You are correct that the ASTM D5291 range for carbon is 75–87, meaning any results outside this range fall outside the method's precision study. While the Charm D5291 results do not fall within the precision limits of the method, we have high confidence in their accuracy, as the area counts remain well within the calibration curve. For sample SR25-2319.001, the area counts were 63,000 and 93,000, with variations due to different weights used (6.6720 mg and 9.8080 mg, respectively). The lowest carbon area count on our calibration curve is 2,400 (0.1350 mg acetanilide), while the highest is 330,000 (19.4930 mg acetanilide). Given this, the area counts of the Charm samples are well within the calibration curve.	Resolved
2	Recommendations	For SGS lab analysis reports, please request and include copies of QA/QC documentation from the lab, if available/applicable, such as any blanks, duplicates, or standards run in conjunction with the C content analyses	WODO Tote samples for PP2 and other COAs	8.2.1.1.1	Provide All lab QA/QC data and reports with analyses	No problem, I will communicate this to our head of analytical chemistry, who owns that relationship	Resolved. Evidence was provided with prior analyses during validation. Strongly recommend providing evidence in all future analytical reports.

3	Clarification	The Emission factor tab in Production Period 4 Summary.xls lists GLEC v3.0 2024 as the source for several factors. The latest version is V3.1 Mar 2025. The factor used is correct (70 kg/t-km) for tankers and flatbeds (85), but text should be edited after reconfirming factors are still correct	Production Period 4 Summary.xls	8.3	Provide clarification and edit file as needed	We update emission factors annually at the beginning of March. So, when that was completed, the March 2025 version of GLEC had not been published yet. The values used were the most recent available when the update was made, which were published in October of 2024.	Understood. Does not impact the calculations as the factors used are correct and updated/unchanged. Resolved.
4	Clarification	Some emission factors now cite GREET 2024 (i.e. LCS)	Production Period 4 Summary.xls	8.3	Please confirm	That's correct -- Activities from March 2025 on cite the GREET 2024 values as part of the annual emission factor update.	Resolved
5	Clarification	On CO Grid Emissions Calc Sheet w 2025 spreadsheet, Solar RECS Calc 2025 tab, cell B1 refers to the 2024 CO grid emissions factor, when I believe it should refer to the 2025 one (?). This does not impact the emissions materially.	CO Grid Emissions Calc Sheet w 2025.xls	8.3	Please confirm	I believe this issue was caught and corrected -- I will provided an updated set of calc sheets for reference. In any case, the purpose of the grid emissions value used in that sheet was to estimate the emissions/net CDR impact of the RECs, so agreed with your assessment that it does not impact emissions calculation since the grid emission factor wasn't used	Resolved
6	Clarification	CO Grid Emissions Calc sheet shows an EF for Solar RECs of 0.035302 kg.kwh for 2025. Production Period 4 Summary shows a EF for Solar RECS of 0.036749.	CO Grid Emissions Calc Sheet w 2025.xls	8.3	Please clarify	I think this may be an issue with the clarity of information presented. The .035302 value is the emission factor without derating. The higher .036749 value is with the derating applied to account for grid losses, which was used in the actual emissions calculation as required by the Energy Use Accounting module Section 5.4	Resolved
7	Clarification	Just wanted to confirm that the salt deliveries are still the waste salt that would have been otherwise disposed	Feb_Mar Salt Deliveries Calc Sheet	NA	Please confirm	That is correct -- If purchased salt is used for removals, that will be accounted for separately	Resolved
8	Clarification	The PDD currently displayed on the registry is v1.8. We received version 1.9 in the last audit. Please work with Isometric to update to proper version.	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx	NA	Please work with Isometric to correct and update	No problem -- This has been raised with Isometric's tech team	Acceptable - confirm at next verification

9	Forward Action Request	There have been some instances of significantly higher C content than typically previously seen (>50%) in some batches now. This may just be a sign of process improvements and achieving higher C content. However, please review batch carbon content history to ensure these are not outliers that could be the result of sample collection, mixing, or other systemic issues and that they are representative. If such samples are outliers, Charm could employ the outlier requirements in Section 8..2.1.1.1 of Protocol to address this as a conservative approach	Charm Blend Batch C Content	8.2.1.1.1	Please provide additional information at next verification	Thank you for raising this, we will review these results.	Acceptable - review at next verification
10	Clarification	For the new Charm oxidizer emissions stack test, can you provide supporting information demonstrating that Oasis is a qualified emissions testing company, accredited to the Stack Testing Accreditation Council for the relevant tests, ISO 17025 accredited, or approved by state regulatory authority to perform compliance emissions testing. Note that this is a strong recommendation, not requirement as currently written	Charm Industrial_Ft Lupton_V1-10 Enclosed Combustor_10-17-24	8.3.6	Please provide additional documentation	The emissions testing company has been contacted and has committed to provide this documentation by COB today. I will forward once received.	Resolved. Review of company website indicates qualifications for state and federal emissions testing compliance. Please provide additional documentation from Oasis when available.
11	Clarification	For the October Stack test data - is there a reason the 2nd test run data was not used in the Emissions Calculations - Methane and N2O spreadsheet? Also, I believe I fully understand the spreadsheet calculations, however, some more annotation and explanation would be helpful for future verifiers. Also, I recalculated the values a different way and get a different value for total kg/day of CO2e. However, the value is still so small that it is not a material impact.	Charm Industrial_Ft Lupton_V1-10 Enclosed Combustor_10-17-24	8.3.6	Please provide additional details regarding emissions factor calculation from stack test data	I can't comment on the specific decisions that went into the calculation as it was not built out by me, but by Grace Connors our head of Engineering. She is currently on maternity leave -- However, the discrepancy between old and new results is well below materiality, accounting for .11% of total reporting period removals	Resolved. Additional clarification would be helpful regarding the calculations when available. As commented, it does not impact claimed removals materially.

12	Clarification	Production Period Summary spreadsheets uploaded on Certify do not match sheets uploaded on Google Drive. For example, PP4 kg emissions/kg material is 0.603 for WODO (as appears in Certify calculations), but supporting evidence file provides a value of 0.492	Certify Platform and Production Period 4 Summary.xls	NA	Please confirm correct data set and update files on Certify	The sheets on Google Drive are correct, and should match the carbon intensity associated with the oil origins used. Old docs contained an error due to the "propane left in tank" from the one production period not being rolled into the total propane consumption in the subsequent production period. That found and corrected after the origins had already been created; we manually updated the carbon intensities so the net removal would be correct, but it looks like the document wasn't updated. I can ping our/Isometric's tech team regarding updated docs	Resolved. Confirmed documents updated on Isometric Certify
13	Forward Action Request	Please provide a copy of the next quarterly Report to KDHE submitted by Vaulted for Q12025 (emplaced material level in cavern). This is not required for the current reporting period, as it would be due in the next reporting period, with a due date of April 28.	NA	9	Submit report at next verification		No action required. Please address at next verification
14	Clarification	Please provide a copy of the disposal well permit for Ajs Services where the salt-water waste was disposed from tank cleanings	Verifier Guidance_ Feb_Mar 2025	NA	Provide copy of permit		Acceptable. Impacts of disposal are not material. Documentation should be provided when available
15	Recommendation (Prior)	To improve stakeholder outreach and community engagement documentation, provide documentation of attendance lists for events as well as lists of questions and answers to demonstrate the level of engagement			Provide additional supporting information if possible	Thank you for the feedback -- This has been communicated to our Community Relations manager	Not Required. Provide updates if any available.
16	Recommendation (Prior)	Provide an improved mechanism for community complaints or inquiries. For example - provide a specific link on the Contact Us or Community Benefits web			Provide additional supporting information if possible	Thank you for the feedback -- This has been communicated to our Community Relations manager	Not Required. Provide updates if any available.

		pages for 'Inquiries or Complaints from the community' or similar					
17	Forward Action Request (prior)	Please provide a copy of the October 24 emissions control burner test when used in future removals accounting for tail gas emissions				Confirmed, that will be included in the submission the subsequent Production Periods	Completed. Report provided in current RP
18	Forward Action Request (Prior)	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1		Please provide AECN testing report. Note, likely not applicable for this reporting period, but should be updated when possible		Not Required. Provide updates in future verifications if available.

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	Criteria Met?	Evidence / Notes *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Kelly Inder-Nesbitt
Senior Carbon Removal Verification Engineer, 350Solutions Inc

Education:

- Master of Science in Geography, Archaeology, and Environmental Studies, University of the Witwatersrand, 2014
- Bachelor of Science with Honors in Geography, University of the Witwatersrand, 2011
- Bachelor of Arts in Geography and Archaeology, University of the Witwatersrand, 2010

Experience Summary:
At 350Solutions, Kelly specializes in verifying carbon removal projects to ensure compliance with ISO 14034 standards and carbon registry requirements. With over a decade of experience in environmental compliance and carbon management, she brings extensive expertise in operational compliance and MRV framework implementation, enhancing accuracy, transparency and integrity in the voluntary carbon market.

Kelly's career spans multiple sectors, including aquaculture, mining, and carbon removal technology, where she has developed and audited environmental management systems that promote sustainable practices and attract investor finance. At 350Solutions, she leads the validation of diverse carbon removal pathways, including biochar, BECCS, DAC and direct ocean capture and biomass burial. Her responsibilities encompass site audits and rigorous evaluation of MRV systems to ensure scientifically validated project claims.

Previously Kelly led the development of Brilliant Planet's carbon dioxide removal methodology protocol for algal biomass burial and contributed as an author. She was also responsible for developing and implementing an ISO 14001 compliant EHSS Management System for the FirstWave Group, who are aquaculture industry leaders in Southern and Eastern Africa. This system is also aligned with IFC World Bank Best Practices and leveraged software tools to streamline compliance monitoring and enhance ESG reporting for investor and regulatory alignment.

Throughout her career, Kelly has consistently collaborated with project developers, communities, regulators, and clients to enhance the credibility of environmental initiatives through rigorous documentation and alignment with international standards. Her approach emphasizes precise data management and actionable reporting, elevating compliance practices into a strategic, value-adding process that drives sustainable business growth.

Kelly's strong communication skills and commitment to fostering collaboration enable her to manage complex compliance initiatives effectively. Her ability to bridge the gap between technical requirements and stakeholder expectations continues to advance science-driven, impactful solutions in the carbon removal industry.